



Java Institute For Advanced Technology

Department of Examinations



COURSE(S) – (LEADING TO)	BSC (HONS) IN SOFTWARE ENGINEERING
UNIT NAME	HANDHELD DEVICE PROGRAMMING II
UNIT ID	JIAT/HHDPII
EXAMINATION ID	JIAT/HHDPII/EX01

Problem Statement

You are expected to develop an M-commerce application as an individual project, formulating a solution for a real-world problem and submit a Project Proposal document prior to commencing the project. This proposal must be approved by the assessor to ensure that the scope meets all necessary requirements to proceed with the project.

Your project should include the following functionalities:

- UI Designing
- Event Handling and Animations
- Data storage (Shared Preferences, SQLite, Internal Storage)
- Network Connection (HTTP Connection)
- Notifications Broadcast
- Receivers and Integration
- Multitasking
- Advanced Components
- Telephony
- Multimedia
- Sensors Controlling
- Google Map (Generic Map, Direction API, Etc.)

Project flow guidelines

01) Project approval

Step 1: Choose the project topic, don't begin by thinking about what to do, but rather, ask yourself the question, "What do I want to know?"

Step 2: If you lack ideas, or wish to gain focus, try some or all of the following strategies:

- Review your course readings, particularly the suggested readings, for topic ideas. Don't just review what you've already read but jump ahead in the syllabus to readings that have not been covered yet.
- Search for real-world problems via the internet, recent introductory books, and if appropriate, more specialized works related to the discipline area of the course.
- Browse through some current journals in your subject discipline. Even if most of the articles are not relevant, you can skim through the contents quickly. You only need one to be the spark that begins the process of wanting to learn more about a topic. Consult a librarian, University online student materials, resources and also contact your tutor about the core journals within your subject discipline.
- Think about essays you have written for past classes and other coursework you have taken or academic lectures and programs you have attended. Thinking back, what most interested you? What would you like to know more about?
- Search online media sources, see if your idea has been covered in the news. Use this coverage to refine your idea

Step 3: Lock your project title

The project title which you submit to your assessor should be able to accurately captures what was done after the research is complete so that the title,

- Indicate accurately the subject and scope of the study.
- Avoid using abbreviations.
- Use words that create a positive impression and stimulate reader interest.
- Use current nomenclature from the field of study.
- Identify key variables, both dependent and independent.
- May reveal how the paper will be organized.
- Suggest a relationship between variables that supports the major hypothesis.
- Is limited to 10 to 15 substantive words.
- Do not include "study of," "analysis of" or similar constructions.
- Titles are usually in the form of a phrase, but can also be in the form of a question.
- Use correct grammar and capitalization with all first words and last words capitalized, including the first word of a subtitle. All nouns, pronouns, verbs, adjectives, and adverbs that appear between the first and last words of the title are also capitalized.

Step 4: Submit your Project Proposal

The Project Proposal should be prepared before you start your project and submitted before the mentioned deadline. Your assessor will then check and approve your Project Proposal, providing you with recommendations (of any). Refer **Appendix 01** for the sample project proposal format.

02) Project Development

Once your Project Proposal is approved by the assessor, you may start developing your project accordingly.

03) Document Submission and Project Viva**Documentation:**

Refer **Appendix 02** for the Final Project Documentation Format.

The Project documentation should present your project work concisely and effectively. It should contain various materials relevant to the work you have undertaken in respect of your project and be organized into a logical framework. It should also be supported by written material that follows well-established academic conventions in a consistent fashion.

Project Viva:

You have to face two main parts at the final viva presentation.

1. Presentation (25 Minutes)

Refer **Appendix 03** for the Guidelines for Presentation & Viva Voce

2. Oral Question session (5 Minutes)

Your assessor may ask a few questions at the end of your presentation, which may seek further clarification on your project. Additionally, there may be technical questions to assess your theoretical knowledge.

APPENDIX 01

PROJECT PROPOSAL FORMAT

Project Name:
XXXXXXXX

Service Type:
XXXXXXXXXXXX

Prepared for:
“CLIENT NAME”

by
“COMPANY NAME”

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1. Document Purpose

This document is meant for “CLIENT NAME” (Hereafter referred to Client)

This copyright of this document rests with “Company Name”. (Hereafter referred to as “Company Name”) and no part of the same should be copied without consulting with the same.

The document outlines our understanding of the project requirements, the objectives of the M-commerce solution, the methodology, deliverables, and commercials.

2. Client Brief

List down all the requirements expected by the client.

3. Objective

Include the following information:

Purpose and Scope of the Application - Define the primary goal of the M-commerce application, such as enhancing online shopping, enabling secure mobile payments, or improving customer engagement.

Key Features and Functionalities - Outline essential features like product browsing, secure payments, user accounts, shopping cart, and order tracking.

User Experience (UX) and Interface Design - Focus on creating a seamless, user-friendly interface that provides an intuitive, efficient, and responsive shopping experience.

Security and Data Protection - Highlight the importance of implementing robust security measures, such as encryption and secure payment gateways, to protect user data and transactions.

Business Impact and Market Advantage - Explain how the application will contribute to the business's growth, enhance customer loyalty, and provide a competitive edge in the market.

4. Scope of Work

Mention in detail the key deliverables that will be provided to the client, including the implementations regarding Functional and Non-Functional Requirements, Design and User Interface, Testing and Quality Assurance etc;

5. Administrative Panel

The Administrative Panel of an M-commerce application provides a comprehensive interface for managing various operational aspects of the platform.

For example, it may include features such as user roles and permissions management, product catalog and inventory control, order processing, customer account management, and payment system configuration. Additionally, the panel may allow access to detailed reports and analytics, security controls like user authentication, and system settings for platform configuration.

6. Standardizations

The Standardizations section should include design, coding, testing, security, performance, documentation, compliance, and delivery standards.

- Design standards ensure responsiveness and cross-browser compatibility, while coding standards emphasize clean, maintainable code.
- Testing standards cover functionality and security checks, and security standards focus on data protection and regulatory compliance.
- Performance standards ensure scalability and optimization, and documentation standards require clear, comprehensive information.
- Compliance standards address legal and accessibility requirements, and delivery standards outline processes for deployment and client handover.

These standards ensure the project meets quality and industry requirements.

7. Delivery Schedule

The Delivery Schedule section outlines key project milestones, including the project kick-off, requirement gathering, design phase with client approval, development, and testing phases. It also includes the deployment, client review, and final delivery of the application, along with post-delivery support. This schedule ensures clear timelines and expectations for each phase, facilitating efficient project completion and client satisfaction.

8. Process Flow

The Process Flow should outline the sequence of steps involved in developing, testing, and delivering the application

9. Requirements from Client

Mention any information/services that is required from the client to develop the project.

Ex: weekly/daily meetings to discuss updates, details like address, telephone numbers, website, photos, product information etc.

10. Location of Work

Mention the location that the project will be developed

11. Progress Reporting and Communication

Mention how the progress of the project will be reported. Ex: formal weekly report, daily conference call etc;

12. Engagement Model

The Engagement Model should include the following areas:

- Intellectual Property: Protection of client information and ownership rights.
- Software Ownership: Transfer of copyright upon full payment.
- Systems & Processes: Project management, workflow, and communication practices.
- Warranty Period and Maintenance: Conditions for bug fixes and changes within the warranty period.
- Terms and Conditions: Scope, client responsibilities, and project delays.
- General Features: Adherence to industry standards and use of open-source solutions.
- Payment Schedule: Payment milestones and conditions for releasing payments.
- Source Code: Transfer of intellectual property and rights to the client upon completion.

Appendix 02

Final Project Documentation Format



Java Institute for Advanced Technology

Department of Examinations

COURSE(S) - LEADING TO	
UNIT NAME	
UNIT ID	
EXAMINATION ID	

NAME	:	
		(BLOCK CAPITALS)
ADMISSION NO	:	
NIC NO	:	

Acknowledgments and Abstract

Being the author of the report, you have to decide the content of this page. This page should be numbered using Roman numeral, i.

Abstract

The abstract of your report should not exceed the maximum of 250 words. This page should be numbered using Roman numeral, ii.

Table of Contents

The table of contents should consist of two levels, i.e., chapter titles as its first level, and section headings as its second level. The third level sub-section headings should not be included in the table of contents

First Level (Chapter title) Include 1. Introduction

Second Level (Section) Include 1.1 Motivation 1.2 Problem Statement

Third Level (Sub-Section) Exclude 1.2.1 Background

List of Tables, Figures, and Abbreviations

The list of tables, figures, and abbreviations should follow the same style as described for the Table-of-Contents page. Long titles should be shortened to precisely explain the respective object. A list of abbreviations should be tied closely in the body of the report, and should not be included if there are less than five abbreviations in the document. The list of tables, figures, and abbreviations page should be numbered Roman numeral.

Captions and Citation Caption

Captions of tables are required to be center-aligned and on the top of the respective table.

Captions of figures are required to be center-aligned and should be below the respective figure.

Citation If you are using a figure, or table from some other sources such as a published article and book, you should cite it in the footnote of the respective object and should be listed in the references section.

Main Body of your Document

1. **Introduction:** A good introduction should tell the reader what the project is about without assuming special knowledge and without introducing any specific material that might obscure the overview. It should anticipate and combine the main points described in more detail in the rest of the project report

2. **Background:** The purpose of the Background section is to provide the typical reader with information that they cannot be expected to know, but which they will need to know in order to fully understand and appreciate the rest of the report. It should explain why the project is addressing the problem described in the report, indicate an awareness of other work relevant to this problem, and show clearly that the problem has not been solved by anyone else.

3. **Specification:** The purpose of the Specification and Design sections is to give the reader a clear picture of the system you plan to create, in terms of the capability required. A specification should tell the reader what the software system is required to do. The design then gives the top-level details of how the software system meets the requirement. It will also identify constraints on the software solution. Describing what a software system does (specification) and how it does so (design) effectively usually means describing it from more than one viewpoint. Each viewpoint will convey some information about the system that other viewpoints omit. (You would use the same technique when describing any complicated construction such as a building, an aircraft, a novel, or a painting)

4. **Design:** A common approach is to first define the user or business requirements, then describe the static architecture, identify modules and groups of closely connected modules, and then apply other views to each of these groups. Fine details, specifically details of code, should be left out. We strongly recommend that you make extensive use of diagrams, such as entity-relationship diagrams, UML diagrams, state charts, or other pictorial techniques (see Section 5.4 for more detailed advice on this).

The design of the system will almost certainly have evolved while you were developing it. Obviously, you should describe its final state but often there are good reasons for describing intermediate states, too; for example, if you want to discuss the details of the design method used or to highlight learning that you later refer to in the Reflection section. If you do this, take special care to make sure the reader does not get confused between different stages of the design. If you are not designing a system, but testing a hypothesis for a more scientifically oriented project, specification, and design sections may not be required in quite the same form.

5. Implementation:

The Implementation section is similar to the Specification and Design section in that it describes the system, but it does so at a finer level of detail, down to the code level. This section is about the realization of the concepts and ideas developed earlier. It can also describe any problems that may have arisen during implementation and how you dealt with them.

6. Results and Evaluation:

In this section, you should describe to what extent you achieved your goals. You should describe how you demonstrated that the system works as intended (or not, as the case may be). Include comprehensible summaries of the results of all critical tests that were carried out. You might not have had the time to carry out any full rigorous tests – you may not even get as far as producing a testable system. However, you should try to indicate how confident you are about whatever you have produced, and also suggest what tests would be required to gain further confidence.

7. Future Work:

It is quite likely that by the end of your project you will not have achieved all that you planned at the start; and in any case, your ideas will have grown during the course of the project beyond what you could hope to do within the available time. The Future Work section is for expressing your unrealized ideas. It is a way of recording that „I have thought about this', and it is also a way of stating what you would like to have done if only you had not run out of time

1. A good Future Work section should provide a starting point for someone else to continue the work which you have begun.

8. Conclusions:

The Conclusions section should be a summary of the aims of the project and a restatement of its main results, i.e., what has been learned and what it has achieved. An effective set of conclusions should not introduce new material. Instead, it should briefly draw out, summarize, combine and reiterate the main points that have been made in the body of the project report and present opinions based on them. The Conclusions section marks the end of the project report properly. Be honest and objective in your conclusions.

9. Reflection on Learning:

Reflection should focus on what Argyris calls “double-loop learning”, this is where we identify, not relatively “simple skills”, such as the mastery of a new programming language, but the impact of what we have done on the assumptions, concepts, and ideas we used to make decisions about our work.

10. References:

You should relate your work to that of other people, but other work explicitly cited should be listed in the Reference section and referred to in the text using some kind of key. It is important that you give proper credit to all work that is not strictly your own, and that you do not violate copyright restrictions. It may be desirable to provide a Bibliography section separately from the reference section. In general, references are those documents/sources cited within the text. The bibliography lists documents that have informed the text or are otherwise relevant but have not been explicitly cited.

Special Note: Plagiarism

Definition Presenting someone else's idea, work, project, or words as your own. This also includes using someone's exact words, figures, tables, et or even modifying them but not providing citation or reference to the actual work.

You can use this characteristic structure as a rough template for organizing the material. However, often it may be of advantage to adjust the suggested structure to your particular project instead of sticking to the template

It is your responsibility to ensure that your project document is formatted correctly before your final submission.

**WARNING:
PLAGIARISM IF PROVED WILL BE GROUNDS FOR DISMISSAL OF
YOUR FINAL PROJECT AND IN WORST CASE DENIAL OF YOUR
DEGREE.**

Appendix 03

Guidelines for Presentation & Viva Voce

GUIDELINES FOR PRESENTATION & VIVA VOCE

A. Guidelines for making Powerpoint slides

Slide layout:

- **PUT** the title at the top of the slide.
- **KEEP** important information near the top of the slide.
- **USE** contrasting colors for text and background.
- **LIGHT** text on a dark background is best. Avoid white backgrounds.
- **KEEP** the color scheme of PPTs consistent throughout your presentation.
- **ADD** one concept per slide.
- **SINGLE** slide should display few key points on it; don't fill up your slides with text in the form of long sentences.
- **USE** bullets to highlight main points and explain these points verbally.
- **AVOID** using frequent animations and transitions in the presentation. Try to apply the same transition, if any, throughout the presentation.
- **USE** clearly visible and understandable diagrams, tables, charts and graphs to enhance the meaning of the text to be presented. It will help in retaining the interest of the audience in the presentation.

Font:

- **CHOOSE** a font that is simple and easy to read such as Arial, Times New Roman or Verdana.
- **USE**, at most, two different fonts – perhaps one for headings and another for content
- **KEEP** all fonts large enough. Font size for main heading should be 40pt, sub headings 32pt & text 24pt.
- **AVOID** using all capital letters – even for headings. ***All Caps*** make the words difficult to read.
- **USE colors on your slides.** Make sure that the color has a purpose: for example, to highlight a part of a figure or to group like things.

B. Guidelines for delivering presentation

- **READ** your document (Internship Report / Project / Dissertation etc.) thoroughly at least 3 to 4 times or until you understand all the details given in it.
- **DECIDE** what is important in your document. A document will have many details, but only a few important ideas. Your presentation should be well planned and focused on the important ideas.
- **START PREPARING EARLY**; don't wait until the last few days to prepare.
- **PRACTICE** your entire presentation-including your slides before a suitable margin of time until you are comfortable. Also, time yourself. Rehearse -- Rehearse and Rehearse. This is the only key of a successful presentation.
- **TRY** to practice before a group of people.
- **THINK ABOUT YOUR AUDIENCE/PANEL**. It is important to anticipate the level of competency the likely audience/panel is expected to have.
- **USE** an effective precise introduction. Try to get familiar with audience/panel.
- **WEAK** introduction should be avoided such as apologies, jokes, symbolic questions.
- **EXPLAIN** to the audience/panel the significance of the topic/issue.
- **ORGANIZE** your presentation clearly and simply: Think logically what should be the ideal and logical sequence of presenting the various concepts/topics covered in the presentation and allocate time accordingly; don't stretch it extra-ordinarily.
- **KEEP** synchronization between the slides and the talk. As you present new ideas, you should be putting up a new slide.
- **USE** body language effectively: Avoid relaxed gestures. Don't play with a pen or pointer.
- **MAINTAIN** eye contact with the audience/panel.
- **MAKE** an effective conclusion: Summarize and provide closure.
- **SHARE** your insight and understanding and tell the audience/panel what you've concluded from your work.